CSTSUNet: A Cross Swin Transformer-Based Siamese U-Shape Network for Change Detection in Remote Sensing Images

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***Abstract*— Change detection (CD) in remote sensing (RS) images is a critical task that has achieved significant success by deep learning. Current networks often employ pixel-based differencing, proportion, classification-based, or feature concatenation methods to represent changes of interest. However, these methods fail to effectively detect the desired changes, as they are highly sensitive to factors such as atmospheric conditions, lighting variations, and phenological variations, resulting in detection errors. Inspired by the transformer structure, we adopt a cross-attention mechanism to more robustly extract feature differences between bitemporal images. The motivation of the method is based on the assumption that if there is no change between image pairs, the semantic features from one temporal image can well be represented by the semantic features from another temporal image. Conversely if there is a change, there are significant reconstruction errors. Therefore, a Cross Swin transformer-based Siamese U-shaped network namely CSTSUNet is proposed for RS CD. CSTSUnet consists of encoder, difference feature extraction, and decoder. The encoder is based on a hierarchical residual network (ResNet) with the Siamese U-net structure, allowing parallel processing of bitemporal images and extraction of multiscale features. The difference feature extraction consists of four difference feature extraction modules that compute difference feature at multiple scales. In this module, Cross Swin transformer is employed in each difference feature extraction module to communicate the information of bitemporal images. The decoder takes in the multiscale difference features as input, injects details and boundaries iteratively level by level, and makes the change map more and more accurate. We conduct experiments on three public datasets, and the experimental results demonstrate that the proposed CSTSUNet outperforms other state-of-the-art methods in terms of both qualitative and quantitative analyses. Our code is available at https://github.com/l7170/CSTSUNet.git.**

***Index Terms*— Change detection (CD), deep learning, remote sensing (RS) image, transformer.**

I. Introduction

CHANGE detection is an important task that involves identifying and analyzing the changes that occur between two or more images and generating change maps. The binary change map consists of pixels with values of either 1 or 0, indicating whether the corresponding position has undergone a change or not. Change detection (CD) technology has a wide range of applications in fields such as urban expansion [1], deforestation [2], damage assessment [3], and other aspects. Accurate CD can help us better understand the dynamics of the environment, detect anomalies and potential hazards, and facilitate decision-making and policy-making.

In the past decade or so, traditional methods for CD in remote sensing (RS) images have mainly relied on image arithmetical-based, image transformation-based, and post classification methods. Image arithmetic-based methods directly compare pixel values from multiple temporal images to generate an image difference map, and then apply a threshold to classify pixels as changed or unchanged. Image transformation-based methods involve transforming the spectral combinations of images into specific feature spaces to differentiate changed pixels. Principal component analysis (PCA) [4], [5] is one of the most widely used algorithms for image transformation and dimensionality reduction. In post-classification methods, the two temporal images are independently classified and labeled, and change areas are extracted by directly comparing the classification results. While these traditional methods have shown significant advancements in CD, arithmetic-based and transformation-based methods heavily rely on empirically designed discriminative feature extraction algorithms and often fail to produce satisfactory results on high-resolution images. Moreover, post-classification methods are highly sensitive to the classification outcomes. The rich details and complex texture features conveyed in high-resolution images rise new challenges for CD tasks. This has led to the emergence of CD methods based on deep learning.

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In deep learning-based CD methods, difference images or change vectors are generated by deep feature comparison. Deep feature-based methods transform bitemporal images into high-level spaces and take deep features as analysis unit. These methods integrate feature extraction and difference discrimination operations within the networks to directly produce the final change maps in an end-to-end manner. Daudt et al. [6] introduced three distinct fully convolutional neural network (FCNN) architectures for CD: the FC-EF architecture, the FC-Siam-conc architecture, and the FC-Siam-diff architecture, all of which could be trained end-to-end. To identify the change of interest in high-resolution RS images, context modeling within the spatial and temporal scope is crucial. Therefore, many studies have been dedicated to enhancing the receptive field of models. On the one hand, attention models [7], [8], [9], [10], [11], [12], [13] have been proposed to capture spatial and temporal dependencies within image pairs for CD tasks. These models leverage an attention mechanism to focus on crucial areas, enabling them to better identify subtle changes in the scene and distinguish them from usual scene variability. On the other hand, transformer models [14], [15], [16], [17], [18], [19] employ a self-attention mechanism to learn global relationships between the image pixels, allowing them to capture long-term dependencies and spatial correlations. Transformer models have achieved impressive performance by utilizing self-attention to model the semantic relationships between any pairs of pixels in the spatiotemporal domain. However, existing transformer-based methods have shortcomings. They either apply self-attention individually to each temporal image to enhance its features or simply use self-attention to reweight the fused bitemporal features/images. Such practices limit the information exchange between the bitemporal image pairs and fail to effectively capture differential information.

To enable mutual learning between the image pairs and better discover differences through information exchange, we propose a Cross Swin transformer-based Siamese U-shaped network, namely CSTSUNet, for RS CD. The interested changes can be represented by several visual tokens, known as semantic markers. The intuition is that if the corresponding tokens between the image pairs do not change, one image token can be expressed by the other image token through cross-attention between the bitemporal images, resulting in a smaller reconstruction error. Conversely, if the corresponding tokens between the image pairs change, one image token cannot effectively be expressed by the other image token, leading to a larger reconstruction error. Therefore, we introduce cross-attention mechanism to facilitate information exchange between the bitemporal image pairs. The change feature is obtained by taking the absolute value of the difference between the original feature maps of bitemporal images and the reconstructed feature maps. This approach differs from previous methods of CD in the way it generates differential information. Moreover, to make cross-attention more efficient when dealing with large-sized images, we utilize the Swin transformer block to develop the Cross Swin transformer. We combine the Swin transformer block with our cross-attention to address this issue. The Cross Swin transformer replaces the self-attention mechanism with the cross-attention mechanism, based on Swin transformer blocks.

In the proposed CSTSUNet, the input bitemporal images are first passed through a feature extraction module to extract multiscale features. Then, the Cross Swin transformer is applied to exchange information between the temporal images at multiple scales. Specifically, for each scale feature, the Cross Swin transformer takes the output feature vectors of one temporal image as input and interacts it with the output feature vectors of the other temporal image. During information exchange, the Cross Swin transformer can help different modules better understand each other's outputs and use this information to improve their own accuracy. Next, the decoder fuses multilevel difference features extracted by the Cross Swin transformer, injects details and boundaries iteratively level by level. Finally, a change map is derived with the same size as the input image.

The contribution of our work can be summarized as follows.

1) To better extract differential features between images, a method called Cross Swin transformer is proposed, which interacts with feature maps from two temporal branches at the same spatial resolution. This interaction motivates each branch to glimpse the representation of the other while preserving its own features. Through this interaction, the Cross Swin transformer can better capture differential features between the two temporal branches.
2) In our model, we adopt the Unet architecture but make a modification that deviates from the Unet structure. Specifically, we incorporate Cross Swin transformer at each skip connection of the Unet to obtain multiscale differential feature maps.
3) The results of extensive experiments on three typical datasets in bitemporal RS image CD demonstrate that the proposed CSTSUNet consistently outperforms other state-of-the-art methods.

The article is structured as follows: Section II provides an overview of related CD algorithms. Section III describes the overall framework of CSTSUNet and explains each component in detail. In Section IV, experimental comparisons are made between CSTSUNet and other state-of-the-art algorithms to demonstrate its effectiveness. Finally, Section V summarizes the performance of CSTSUNet.

II. Related Work

RS CD has been an active research area for decades, and various methods have been proposed to tackle this problem. Early CD methods include pixel-based differencing, ratioing, and classification-based methods. While these methods have achieved some success, they are often limited by their sensitivity to changes in atmospheric and illumination conditions, making them less effective for detecting subtle changes.

Recently, deep learning-based methods [20], [21], [22] and transformer-based methods [16], [23], [24] have been proposed for bitemporal RS image CD. Deep learning-based methods typically use convolutional neural networks (CNNs) to extract features from the input images. These features are then fed into a classification or segmentation model that detects changes

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between the two images. Transformer-based methods, on the other hand, use attention mechanisms to selectively focus on relevant parts of the input images and extract features from them. These methods have shown great potential for improving CD performance.

A. CNNs Methods

CNNs have been widely applied to CD in RS imagery. CNNs are a type of deep learning architecture that have shown outstanding performance in various computer vision tasks, including image classification [25], [26], [27], object detection [28], [29], [30], and segmentation [31], [32], [33]. In CD, CNNs can be employed to extract high-level features from images, which can be used to identify subtle changes that are imperceptible to the human eye. There are several famous CNN architectures that have been used for CD, including the fully convolutional network (FCN) [34], the U-Net [35], and the residual network (ResNet) [36]. These architectures differ in their structure and design, but share the common characteristic of using convolutional layers to extract multiple features.

The U-Net architecture is a popular choice for CD tasks, which is an encoder–decoder architecture that has been shown to achieve the state-of-the-art performance in various segmentation tasks. The U-Net consists of an encoder network that downsamples the input images and an upsampling decoder network that generates the output segmentation map. Due to a pair of images as input in bi-temporal CD, siamese [37] U-net with the share weight parameters is applied to extract the multiscale features. Two outputs of Siamese U-net are concatenated to the decoder network. Finally, the change map can be derived by the decoder network. Another popular CNN architecture for CD is the ResNet, which employs skip connection to avoid the problem of vanishing gradients during training. The skip connection enables the model to learn the residual function, which is the difference between the input and output of each layer, and has been proven to improve the performance effectively [38].

Siamese convolutional networks and end-to-end training methods has become prevalent in CD tasks, leading to significant improvements in the accuracy of CD. For example, Zhan et al. [39] were the first to propose a Siamese convolutional network for CD task. This structure allows for simultaneous processing of bitemporal images, and it has become the basis of many current CD methods. The Siamese structure has proven to be an effective approach to CD, as it can effectively learn the difference between bitemporal images. Daudt et al. [6] proposed three different FCNN-based architectures for CD, all of which could be trained end-to-end. The FC-EF architecture concatenated the bitemporal images as input, while the FC-Siam-conc and FC-Siam-diff architectures used a Siamese structure to directly process bitemporal images. These architectures were shown to be effective for CD.

B. Attention-Based Methods

To address the issue of limited receptive fields in convolutional operations, which can lead to a failure to capture global information, various methods have been proposed to increase the receptive field. Among these methods, attention mechanisms have been shown to be effective in improving the performance of deep learning models. In the context of CD, attention-based methods can be used to focus on the most informative parts of the image, such as regions with significant changes. For example, Liu et al. [7] proposed an attention-based CD method that uses a spatial and channel attention mechanism to enhance the discriminative power of features. Fang et al. [8] proposed an attention-based CD method that leverages an ensemble channel attention module (CAM) to fuse features of various levels, enabling the generation of stronger change features. In [8], Siamese architecture and dense connections between feature maps are employed to improve information flow and achieve better performance in detecting changes. The ensemble CAM is designed to enhance the discriminative power of features by fusing information from different levels, and achieved the state-of-the-art results on benchmark datasets.

C. Multiscale Methods

Multiscale analysis is an important approach for CD tasks, and it can capture changes at different scales by analyzing the image at multiple resolutions. For example, Zhang and Shi [40] proposed a feature difference CNN-based CD method that incorporates multiscale information in feature extraction. The method uses a multiscale feature difference network that extracts features at multiple scales and fuses them to detect changes. The multiscale feature difference network includes multiscale filters and parallel feature extraction branches to extract and combine features from different scales. Bandara and Patel [41] proposed a transformer-based Siamese network for CD that utilizes multiscale features. The method uses a Siamese network that processes images at different scales and extracts features using a transformer-based architecture. The method also employs attention mechanisms to focus on important features and reduce computation time. Chen et al. [42] proposed a deep Siamese multiscale convolutional network for CD in multitemporal very high-resolution (VHR) images. The method uses a Siamese network that processes images at different scales and extracts features using a multiscale convolutional network. The multiscale convolutional network includes multiple convolutional layers that extract features at different scales and a fusion layer that combines features from different scales.

D. Transformer-Based Methods

In recent years, a new type of neural network architecture called the transformer [43] has been introduced to the computer vision field, initially for natural language processing, and later on for image-related tasks as well. The vision transformer (ViT) [44] is the first pure transformer approach for image classification tasks, where 2-D image patches with positional embedding are used as inputs and the network is pretrained on large datasets. ViT has achieved comparable performance with CNN-based methods in terms of image classification accuracy. Another recent approach, the hierarchical

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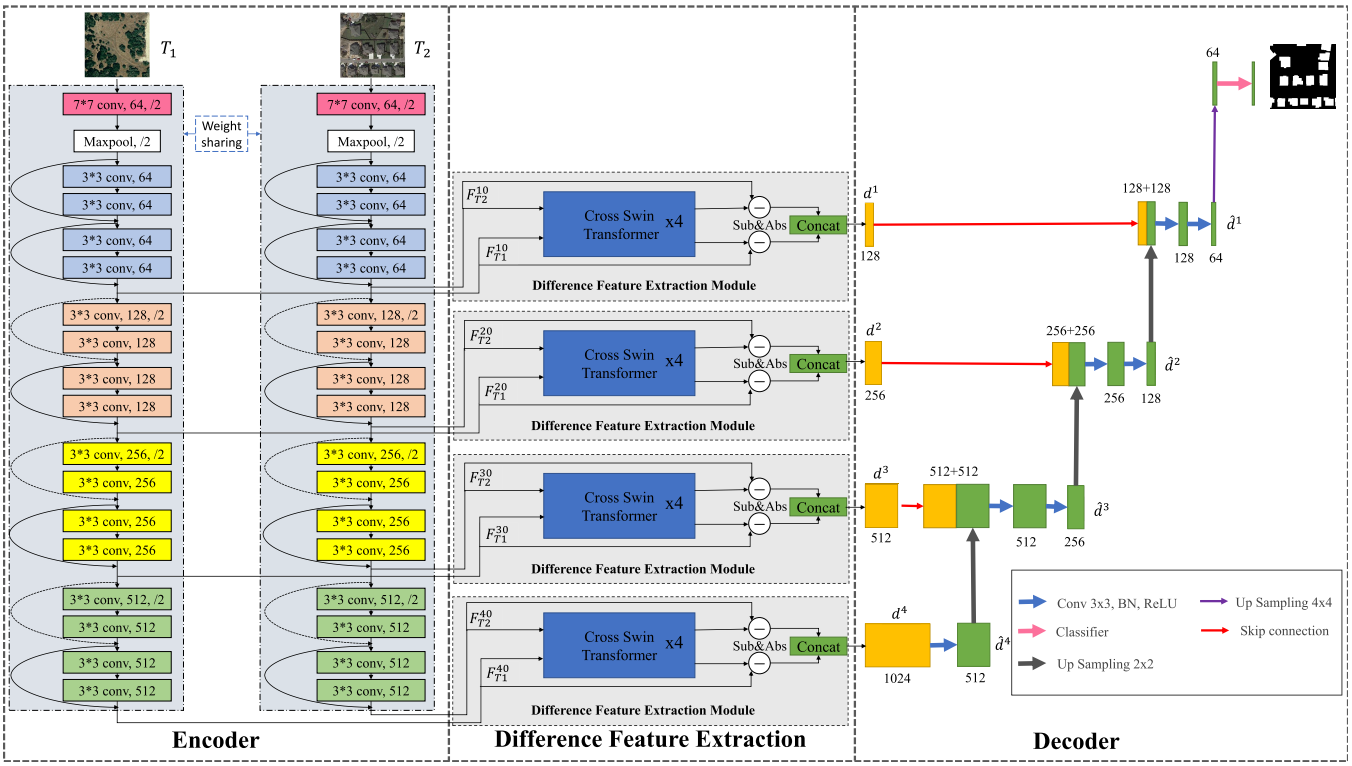


Fig. 1. Framework overview of CSTSUnet. CSTSUnet extracts multiscale features from paired time-lapse images using an encoder, followed by a difference feature extraction module that extracts difference features, and finally generates a change map using a decoder. In this illustration, ResNet18 is utilized as the encoder.

Swin Transformer [45], uses a shifted windowing scheme to calculate self-attention in a local window, thereby reducing the computational complexity of self-attention. In the context of CD, SwinSUNet [14] is a method that uses a pure transformer network to extract features and perform CD. It decomposes the input image into multiple blocks and performs feature extraction and CD using a transformer network. Additionally, researchers have also combined transformers with CNNs for CD. Chen et al. [19] proposed a bitemporal image transformer (BiT) that uses CNN to generate semantic features and leverages a transformer module to further process CNN features, achieving the state-of-the-art results in CD. Moreover, ICIF-Net [15] uses an intrascale cross-interaction and interscale feature fusion network to extract high-level features using a transformer network, which are then fused with low-level features extracted by a CNN. This approach has shown improved performance in CD tasks.

The application of CNNs, attention-based methods, multiscale techniques, and transformer-based approaches in CD has significantly improved the performance of CD. The proposed CSTSUNet shares similarities with these approaches while also introducing novel elements. In the proposed method, the encoder is based on a hierarchical ResNet with the Siamese U-net structure. Then, the Cross Swin transformer is applied to exchange information between the temporal images at multiple scales. Finally, using convolution network, the decoder gradually restores the change information in the image. The proposed method differs from previous CD methods in the way it generates difference information. We introduce a cross-attention mechanism to facilitate information exchange between the bi-temporal image pairs. By computing the absolute difference between the original feature maps and the reconstructed feature maps of the bitemporal images, we obtain the change features. Furthermore, to make the cross-attention more effective in handling large-sized images, we leverage the Swin transformer block to develop the Cross Swin transformer. The Cross Swin transformer is based on the Swin transformer block and replaces self-attention with the cross-attention mechanism. The combination of the Swin Transformer block and cross-attention effectively addresses this issue. The proposed method demonstrates significant performance improvement in CD tasks.

III. Proposed CD Framework

A. Framework Overview

As illustrated in Fig. 1, Unet structure is employed in our proposed CSTSUnet. In CSTSUnet, we utilizes ResNet as encoder of Unet to extract multiscale feature maps. Unlike traditional Unet structures, skip connection is replaced with the Cross Swin Transformer to extract the difference features of bitemporal images. This allows the network to capture multiscale difference features, which can be decoded into difference images. The combination of Unet and Cross Swin transformer enables the network to effectively extract difference features from dual-time images and produce more accurate difference images.

Specifically, our proposed CSTSUnet comprises three primary components. The first component is a hierarchical convolution encoder based on the ResNet network, which

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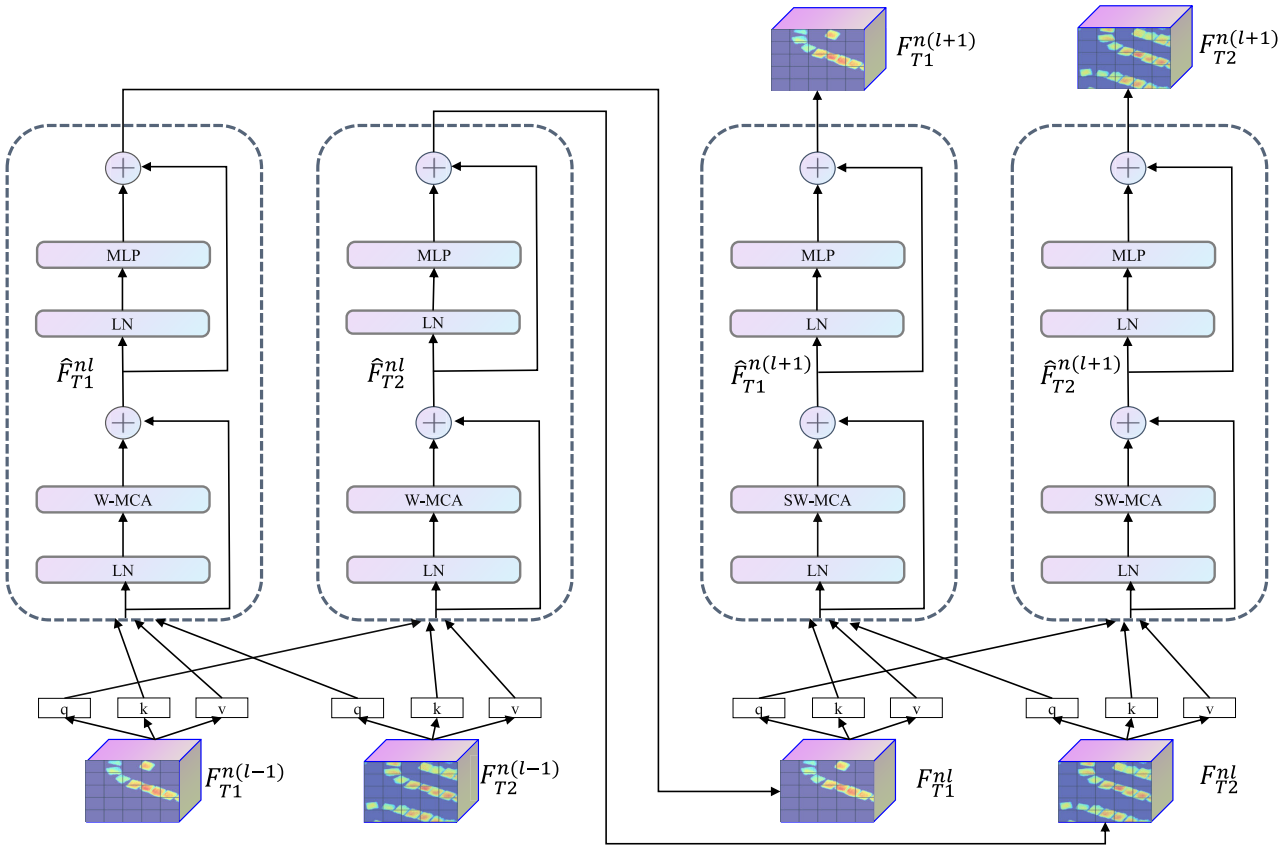


Fig. 2. Two successive cross Swin transformer, including W-MCA module and SW-MCA module.

is integrated into a Siamese network to extract both coarse and fine features from bitemporal images. In this module, the multiscale feature maps of each input image are obtained separately by convolutions. The second component consists of four difference feature extraction modules that compute difference feature at multiple scales. In this module, Cross Swin transformer is employed in each difference feature extraction module to communicate the information of bitemporal images. Then, the decoder is applied to fuse these multilevel difference features which is important to generate accurate change maps. Finally, the linear projection is employed to predict the change map.

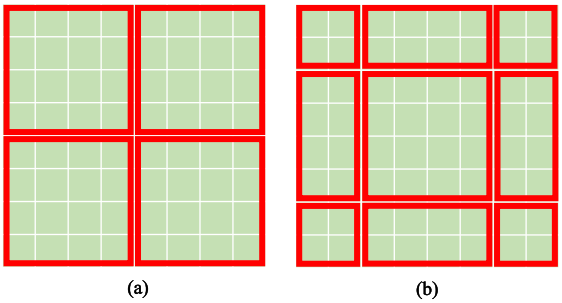


Fig. 3. Window partition strategy of (a) W-MCA module and (b) SW-MCA module.

B. Cross Swin Transformer

The framework of Cross Swin transformer is shown in Fig. 2. We propose Cross Swin transformer-based Swin transformer block. The key distinction between the Swin transformer block and our approach is that (shifted) window-based multihead self-attention [(S)W-MSA] module is replaced by (shifted) window-based multihead cross-attention [(S)W-MCA] module, enabling cross-attention between two temporal images instead of self-attention on a single image. And both of them includes multilayer perceptron (MLP) module, LayerNorm (LN) layers, and residual connections.

1) Window-Based Multihead Cross-Attention (W-MCA) Module: W-MCA stands for window-based multihead cross-attention, which is an attention mechanism used for processing image data. The W-MCA module divides the feature map into several windows of size $M \times M$, as shown in Fig. 3(a). This module calculates cross-attention separately in each window. The formula for calculating cross-attention in the W-MCA module is the different from in the standard window-based multihead self-attention (W-MSA) module. In W-MSA, they use the same input vectors to calculate the query, key, and value vectors, and then pass them through the attention mechanism to obtain the output vectors. However, in the W-MCA mechanism, the query vector usually comes from a feature map of one time step, while the key and value vectors come from a feature map of another time step.

In other words, the W-MCA mechanism is able to capture the correlations between two time steps and use information from the previous time step in the calculation of the feature map for the current time step. This approach can better capture

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the spatial and temporal dependencies in the image, thus improving the performance of CD tasks.

2) SW-MCA Module: Based on the SW-MSA module, the SW-MCA module is proposed to process the bitemporal images. We use a window-shift operation in the SW-MCA module to compute attention between adjacent windows, thereby expanding the attention scope on the feature map. Unlike the SW-MSA module, the SW-MCA module uses cross-attention to process two sets of feature maps from different temporal. The SW-MCA module divides the feature maps of the two temporal into several windows of size $M \times M$ and performs cross-attention separately in each window. In each window, the query vector usually comes from a feature map of one temporal, while the key and value vectors come from a feature map of another temporal. The output vectors are then obtained through the cross-attention mechanism.

The window partition strategy of the SW-MCA module is the same as the SW-MSA module, as shown in Fig. 3(b). with a window offset of $(M/2, M/2)$ relative to the previous window. In the Cross Swin transformer, the W-MCA module alternates with the SW-MCA module to increase the network receptive field.

3) Cross-Attention: In CD, cross-attention can be used to process multitemporal RS image data. In this case, each temporal image can be considered as a modality, where the query comes from one temporal image and the keys and values come from another temporal image. Cross-attention allows the model to interact and learn information across different temporal images for better detection of land cover changes

$$\begin{aligned} Q &= F_{Ti}\ W^q \\ K &= F_{Tj}\ W^k \\ V &= F_{Tj}\ W^v \\ \text{Attention}(Q,\ K,\ V) &= \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \end{aligned} \tag{1}$$

where F_{Ti}, $F_{Tj} \in \mathbb{R}^{M^2 \times d}$ represents feature vectors extracted from different temporal images, F_{Ti} is obtained from one temporal image with $i = \{1, 2\}$, while F_{Tj} is obtained from another temporal image with $j = \{2, 1\}$. W^q, W^k, $W^v \in \mathbb{R}^{d \times d}$ represent the weights of three linear projection layers. $Q, K, V \in \mathbb{R}^{M^2 \times d}$ represent the query, key, and value matrices, respectively. M^2 denotes the resolution of F_{Ti}, F_{Tj}, Q, K, and V and d denotes the channel dimension. $B \in \mathbb{R}^{M \times M}$ is a relative position bias matrix, which is a learnable parameter.

4) Cross Swin Transformer: In Fig. 2, two successive Cross Swin transformer are depicted. The first Cross Swin transformer utilizes a W-MCA module, while the second Cross Swin transformer utilizes an SW-MCA module. As previously explained, these modules employ different window partitioning strategies to mitigate the impact of window locality. The calculation equations for the two successive Cross Swin transformer are provided below

$$\begin{aligned} \hat{F}_{Ti}^{nl} &= \text{WMCA}\left(\text{LN}\left(F_{Tj}^{n(l-1)}\right), \text{LN}\left(F_{Ti}^{n(l-1)}\right)\right) + F_{Ti}^{n(l-1)} \\ F_{Ti}^{nl} &= \text{MLP}\left(\text{LN}\left(\hat{F}_{Ti}^{nl}\right)\right) + \hat{F}_{Ti}^{nl} \\ \hat{F}_{Ti}^{n(l+1)} &= \text{SWMCA}\left(\text{LN}\left(F_{Tj}^{nl}\right), \text{LN}\left(F_{Ti}^{nl}\right)\right) + F_{Ti}^{nl} \\ F_{Ti}^{n(l+1)} &= \text{MLP}\left(\text{LN}\left(\hat{F}_{Ti}^{n(l+1)}\right)\right) + \hat{F}_{Ti}^{n(l+1)} \end{aligned} \tag{2}$$

where $\hat{F}_{Ti}^{nl}(i = 1, 2)$ represents the output of (S)W-MCA module, while $F_{Ti}^{nl}(i = 1, 2)$ represented the output of MLP module. Here n stands for the nth layer of the ResNet network, which can have values from 1 to 4. On the other hand, l represents the lth layer of Cross Swin transformer. Each difference feature extraction module employs four Cross Swin transformer.

C. Encoder

The encoder uses a hierarchical ResNet18 network to extract multilevel features from input images. The hierarchical ResNet18 consists of four layers, with each layer comprising of residual blocks. Each residual block in ResNet18 is composed of two convolutional layers, batch normalization (BN) layers, and rectified linear unit (ReLU) activation functions. The feature map resolution is halved and dimension is doubled at each layer. After four layers, the feature resolution is reduced from $(H/4) \times (W/4)$ to $(H/32) \times (W/32)$, while the dimension is increased from C to $8C$. To ensure efficient learning from both bitemporal images, we implement weight sharing between the two image branches in the encoder. Specifically, the weight sharing mechanism involves sharing the parameters (weights and biases) of the corresponding layers in both branches. By sharing weights between the two branches, the model can effectively leverage information from both bitemporal images and learn common patterns and features that are relevant for CD.

At each layer of the encoder, the output feature maps are saved and denoted as $F_{T_1}^{n0}$ and $F_{T_2}^{n0}$, where n refers to the nth layer of ResNet18. These feature maps are then input to the corresponding Cross Swin transformer to extract multiscale differential features. The extracted differential features are subsequently used in combination with decoder features to recover change information. It is easy to see that multiscale features can be generated by encoder, in which the high-level features hold strong semantic information and the low-level features retain the detailed information.

D. Difference Feature Extraction

The difference feature extraction component is responsible for calculating difference features at multiple scales, which is achieved using four difference feature extraction modules. Each module calculates the difference between the feature maps obtained from two input images captured at different times. To facilitate the calculation of accurate difference features, four Cross Swin transformer are used in each difference feature extraction module. This mechanism enables information exchange between the feature maps from the dual-time images, resulting in more accurate and informative difference features. Specifically, we first obtain the feature maps of the dual-time images from the encoder. Then, each feature map at each time point is processed by a Cross Swin transformer to obtain a feature map with stronger difference representation ability. Next, for each time point, the feature map is subtracted from its corresponding enhanced feature

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map containing difference information, and the absolute value is taken to obtain the difference map. In the process of subtraction and absolute value taking, we can highlight significant difference parts, which helps the model better capture changes between different time points. The enhanced feature map is obtained by further processing the original feature map. Finally, we concatenate the difference feature maps of the dual-time images and input the concatenated difference feature map into the encoder module. The difference feature maps obtained at different scales are combined to generate accurate difference maps. The formula for calculating a single differential feature extraction module is as follows:

$$\begin{aligned} F_{Ti}^{n4} &= \mathrm{CST}\left(F_{Ti}^{n0}\right) \\ D_i^n &= \left|F_{T1}^{n4} - F_{T2}^{n0}\right| \\ d^n &= \mathrm{Concat}\left(D_1^n, D_2^n\right) \end{aligned} \tag{3}$$

where $F_{Ti}^{n0}(i = 1, 2)$ represents the feature map of a single temporal image obtained from different layers of the encoder. Here, n stands for the nth layer of the ResNet network, which can take values from 1 to 4. CST stands for Cross Swin transformer. After being processed by CST, the single-time image feature map F_{Ti}^{n0} obtains the feature map F_{Ti}^{n4} that has undergone information exchange between the feature maps from the dual-time images. The difference feature D_i^n of the image is obtained by subtracting and taking the absolute value of F_{T1}^{n0} and F_{T2}^{n4}. Concatenate the difference feature maps D_1^n and D_2^n of the dual-temporal images at the same scale to obtain the difference feature d^n.

E. Decoder

The decoder fuses multilevel difference features, injects details and boundaries iteratively level by level, and predicts the change map. Specifically, using convolution network, the decoder can gradually restore the change information in the image. The upsampling component is responsible for expanding the feature map to the same size as the original input images. This step is important to generate accurate change maps.

In the decoder, the output of the bottom-level difference feature extraction module is processed by a 3×3 convolution to reduce the number of channels by half. Then, in each subsequent step, the feature map is upsampled using bilinear interpolation, connected with the corresponding difference feature map from the difference feature extraction module, and passed through two 3×3 convolutions, each followed by BN and ReLU. Each convolution operation reduces the number of channels by half. This design progressively restores the input image size and number of channels, while integrating feature information from different levels to help generate more accurate output. The process can be represented using the following formulas:

$$\hat{d}^4 = \mathrm{Conv}(\mathrm{d}^4) \tag{4}$$

$$\begin{aligned} \hat{d}^{n-1} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(\mathrm{ReLU} \\ (\mathrm{BN}(\mathrm{Conv}(\mathrm{Concat}(d^{n-1}, \mathrm{Up}(\hat{d}^n)))))))) \end{aligned} \tag{5}$$

where d^{n-1} represents the corresponding difference feature map from the difference feature extraction module. Here n can have values from 2 to 4. Conv, BN, ReLU, Up, and Concat represent the convolution operation, BN, activation function, upsampling method, and merging along channels, respectively.

Next, the difference feature map d^1 obtained earlier is upsampled by a factor of four using linear interpolation, resulting in an upsampled feature map $X \in \mathbb{R}^{H_0 \times W_0 \times C}$, where H_0 and W_0 represent the height and width of the original image, respectively. Then, a change probability map $P \in \mathbb{R}^{H_0 \times W_0 \times 2}$ is generated. The change probability map is calculated as follows:

$$P = \sigma\left(g\left(\mathrm{Up4}\left(d^1\right)\right)\right) \tag{6}$$

where Up4 represents the process of upsampling the feature map by a factor of four. the change classifier, denoted as $g : \mathbb{R}^{H_0 \times W_0 \times C} \rightarrow \mathbb{R}^{H_0 \times W_0 \times 2}$, is responsible for predicting changes in the upsampled feature map. The classifier consists of two 3×3 convolutional layers with BN applied. Each convolutional layer has output channels of "64, 2," respectively. After passing the upsampled feature map through the change classifier, a softmax function, denoted as $\sigma(.)$, is applied pixel-wise on the channel dimension of the classifier's output. This softmax operation calculates the probabilities of change and no change at each pixel location, resulting in the final change probability map with dimensions $\mathbb{R}^{H_0 \times W_0 \times 2}$.

During the inference phase, the prediction mask, $M \in \mathbb{R}^{H_0 \times W_0}$, is computed based on the change probability map, P. The prediction mask is obtained by performing a pixel-wise Argmax operation along the channel dimension of the change probability map. This operation selects the channel with the highest probability at each pixel location, resulting in a mask.

F. Loss Function

In the training stage, we minimize the cross-entropy loss to optimize the network parameters. For- mally, the loss function is defined as

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}) \tag{7}$$

where $l(P_{hw}, y) = -\log(P_{hwy})$ is the cross-entropy loss, and Y_{hw} is the label for the pixel at location (h, w).

IV. Experiments

A. Datasets

We conduct experiments on three CD datasets.

The learning, vision, and RS (LEVIR)-CD dataset [46] is a public available dataset designed for building CD in RS imagery. It consists of 637 pairs of high-resolution (0.5 m) RS images, each with a size of 1024×1024 pixels. The dataset is split into training, validation, and test sets based on the default dataset split. To overcome GPU memory limitations, the images are divided into small non-overlapping patches of size 256×256 pixels, resulting in a total of 7120, 1024, and 2048 pairs of patches for the training, validation, and test sets, respectively.

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The CDD dataset [47] is a collection of bitemporal images that includes 11 pairs with seasonal changes. These images were obtained from Google Earth and have spatial resolutions ranging from 3 to 100 cm/pixel. The dataset comprises seven pairs of images sized at 4725 × 2200 pixels and four pairs of images sized at 1900 × 1000 pixels. The changes that need to be detected in the CDD dataset include those caused by buildings, cars, and roads, among others, but not those caused by seasons. In [48], To generate the dataset, all images were partitioned into 256 × 256 image patches by cropping and rotating. In total, 16 000 pairs of patches were created, of which 10 000 pairs were used for training, 3000 pairs for validation, and the remaining patches for testing.

The deeply supervised image fusion network with CD (DSIFN-CD) [49] is a publicly available binary CD dataset consisting of six pairs of high-resolution (2 m) satellite images from six major cities in China. The dataset includes changes in various types of land cover objects, such as roads, buildings, croplands, and water bodies. The bi-temporal images of Beijing, Chengdu, Shenzhen, Chongqing, and Wuhan have been clipped into 394 sub-image pairs with dimensions of 512 × 512. After data augmentation, a total of 3940 bi-temporal image pairs have been acquired. Additionally, the Xian image pair has been clipped into 48 image pairs. We then crop non-overlapping small blocks of size 256 × 256 and randomly divide them into three parts in a ratio of 8:1:1 to create training/validation/testing sets with sample sizes of 12672/1595/1596, respectively.

B. Evaluation Criteria

In our scientific research, we utilized several evaluation metrics to assess the accuracy of our experimental results. These metrics included true positive (TP), true negative (TN), false positive (FP), false negative (FN), precision, recall, $F1$-score, intersection over union (IoU) of the change category, and overall accuracy (OA). TP, FP, FN, and TN represent the number of true positives, false positives, false negatives, and true negatives, respectively. Among these metrics, $F1$-score was identified as the most representative and effective in reflecting the quality of the experimental results.

The calculation formulas of precision, recall, $F1$-score, and OA are given as follows:

$$\begin{aligned} \text{precision} &= \text{TP}/(\text{TP}+\text{FP}) \\ \text{recall} &= \text{TP}/(\text{FP}+\text{FN}) \\ F1 &= \frac{2}{\text{precision}^{-1}+\text{recall}^{-1}} \\ \text{OA} &= (\text{TP}+\text{TN})/(\text{TP}+\text{TN}+\text{FP}+\text{FN}). \end{aligned} \tag{8}$$

C. Implementation Details

In our experiments, we used a ResNet model pretrained on ImageNet to initialize the parameters of CSTSUnet. This pretrained model is provided in [36]. Then, we fine-tuned the model on the CD dataset to further optimize the parameters.

We implemented our model using the PyTorch framework and trained, validated, and tested it on an NVIDIA GeForce RTX 3090 graphics card. We set the patch size to 8 and used the SGD optimizer with an initial learning rate of 0.01, a momentum of 0.99, and a weight decay of 0.0005. We trained the model for 200 epochs, with a linear decay strategy that gradually reduced the learning rate from 0.01 to 0 over the 200 epochs to improve the training performance. We applied simple data augmentation techniques to the input image patches, including flipping, rescaling, cropping, and Gaussian blurring. After each epoch, we validated the model using a validation set and saved the model with the highest validation accuracy. We used the model parameters with the highest final accuracy for testing.

D. Comparative Experiments

1) Comparative Methods: we conducted a comparative experiment to evaluate the performance of several state-of-the-art methods for CD in RS images. The methods we compared included three pure convolution-based methods: early fusion (FC-EF) [6], fully convolutional Siamese-difference (FC-Siam-Di) [6], and fully convolutional-Siamese-concatenation (FC-Siam-Conc) [6], as well as two attention-based methods: dual task constrained deep Siamese convolutional network (DTCDSCN) [7] and SNUNet [8]. There was also a method that combined CNN and transformer: BIT [19].

a) FC-EF [6]: FC-EF is a pure convolution-based method for CD in RS images. It involves fusing the two input images, representing the prechange and post-change scenes, at an early stage of the network using a simple concatenation operation. The concatenated image is then passed through a deep CNN for CD. FC-EF is known for its simplicity and speed, making it an attractive option for real-time applications.

b) FC-Siam-Conc [6]: FC-Siam-Conc is another pure convolution-based method for CD. It involves constructing a fully convolutional Siamese network, where the prechange and post-change images are passed through two identical CNNs with shared weights. The output feature maps are then concatenated and fed through a few additional convolutional layers to generate the change map. FC-Siam-Conc is designed to exploit the complementary information between the two input images, leading to improved CD performance.

c) FC-Siam-Di [6]: FC-Siam-Di is similar to FC-Siam-Conc in that it uses a fully convolutional Siamese network. However, instead of concatenating the feature maps, FC-Siam-Di computes the absolute difference between the feature maps from the prechange and post-change images. The difference map is then passed through additional convolutional layers to generate the final change map. FC-Siam-Di is known for its effectiveness in capturing subtle changes in RS images.

d) DTCDSCN [7]: DTCDSCN is an attention-based method that uses a dual-task framework to learn both a CD task and a change localization task simultaneously. The method involves constructing a deep Siamese network with a shared encoder and two independent decoders, each supervised by the label maps of the corresponding temporal. However, for the sake of fairness in comparison, the semantic segmentation decoders were omitted from the evaluation.

e) SNUNet [8]: SNUNet is another attention-based method for CD in RS images. It involves constructing a U-shaped network with skip connections to capture multiscale

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TABLE I
COMPARISON RESULTS ON THE LEVIR TEST SET. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). COLOR CONVENTION: **BEST**, **SECOND-BEST**

	Pre	Rec	F1	IoU	OA
FC-EF [6]	88.06	83.70	85.82	75.17	98.59
FC-Siam-Conc [6])	91.57	86.04	88.72	79.73	98.89
FC-Siam-Di [6]	**92.34**	83.23	87.55	77.85	98.79
DTCDSCN [7]	72.99	**96.29**	83.03	70.99	98.00
SNUNet [8]	89.50	88.87	89.18	80.47	98.90
BIT [19]	89.24	89.37	**89.31**	**80.68**	**98.92**
CSTSUNet	**91.99**	**89.41**	**90.68**	**82.96**	**99.06**

TABLE II
COMPARISON RESULTS ON THE CDD TEST SET. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). COLOR CONVENTION: **BEST**, **SECOND-BEST**

	Pre	Rec	F1	IoU	OA
FC-EF [6]	84.74	66.29	74.39	59.22	94.61
FC-Siam-Conc [6])	92.08	69.67	79.32	65.73	95.71
FC-Siam-Di [6]	90.29	65.52	75.26	60.34	95.00
DTCDSCN [7]	87.27	**98.17**	92.40	85.87	98.09
SNUNet [8]	95.68	94.66	95.17	90.78	98.87
BIT [19]	**96.03**	94.32	**95.17**	**90.78**	**98.87**
CSTSUNet	**96.72**	**96.56**	**96.64**	**93.50**	**99.20**

information. The method uses a spatial and channel attention mechanism to learn the relevance of each pixel in the image, allowing it to focus on the most informative regions for CD. SNUNet is known for its high accuracy and efficiency.

f) BIT [19]: BIT is a method that combines CNN and Transformer for CD in RS images. It involves using a CNN to extract features from the input images, followed by a ransformer module to learn the long-range dependencies between the features. The resulting feature maps are then passed through additional convolutional layers to generate the change map. BIT is known for its ability to capture both local and global changes in RS images.

In these comparative experiments, we implemented the model parts of FC-EF [6], FC-Siam-Di [6], and FC-Siam-Conc [6] using publicly available code online. For all three methods, we used cross-entropy as the loss function and trained the models using the SGD optimizer with an initial learning rate of 0.01. For DTCDSCN [7], SNUNet [8], and BIT [19], we implemented these CD networks using their publicly available codes with default hyperparameters.

2) Experimental Results on LEVIR Dataset: Table I presents several comparative experiments and the results of our proposed CSTSUNet method on the LEVIR dataset. The best and second-best results of the metrics are highlighted in red and blue, respectively. It can be seen from the table that CSTSUNet achieves the highest scores on three evaluation metrics among all the experimental methods. These evaluation metrics include $F1$ score, IoU, and OA. Specifically, CSTSUNet surpasses the highest score of other experimental methods by 1.37, 2.28, and 0.14 points on $F1$ score, IoU, and OA, respectively.

We have chosen a number of images from the LEVIR test set and applied various algorithms to generate change maps. The visualization comparison of the methods on the LEVIR datasets is displayed in Fig. 4. To facilitate the interpretation of the comparison results, we have used different colors to represent TPs in white, TNs in black, FPs in red, and FNs in green. From these visualization results, it is easy to see that the change maps generated by CSTSUNet have better details and are closer to the ground truth than other methods.

The results from our approach demonstrate excellent CD capabilities for densely changed buildings, as evident from in Fig. 4(a) and (b). Furthermore, our CSTSUNet is capable of generating relatively complete predictions for larger changed areas, as illustrated in Fig. 4(c) and (d). In comparison, other methods struggle to detect changed areas under strong lighting changes, as shown in Fig. 4(e), whereas our CSTSUNet approach can effectively identify such changes. Overall, these findings suggest that our method is highly effective in detecting changes, particularly in scenarios involving dense changes and strong lighting variations.

3) Experimental Results on CDD Dataset: Table II displays the results of our proposed CSTSUNet method and several comparative experiments on the CDD dataset. The table indicates that CSTSUNet outperforms all other experimental methods, achieving the highest scores on four evaluation metrics, including Precision, $F1$ score, IoU, and OA. Notably, our proposed method outperforms the highest score achieved by other experimental methods by 0.69, 1.47, 2.72, and 0.23 points on $F1$ score, IoU, and OA, respectively. These results demonstrate the superior performance of CSTSUNet in CD compared to other experimental methods.

We selected several images from the CDD test set and utilized algorithms identical to those used in the LEVIR dataset to generate change maps. The comparative visualization of these methods on the CDD dataset is presented in in Fig. 5. The results of our visualization clearly show that the change maps generated by CSTSUNet are superior in terms of edge details, and are better suited to handle change maps with interference.

The images in Fig. 5(a)–(c) illustrate changes that occur due to the presence of buildings, with seasonal variations evident in (a) and (b) and occlusion present in (d). However, it is apparent from the images that numerous methods are unable to precisely detect building changes during different seasons or identify changes caused by building occlusion due to tree leaves. In Fig. 5(d), changes caused by the presence of buildings and cars are shown, Fig. 5(e) depicts changes due to the interaction of cars and roads. These images demonstrate that CSTSUNet outperforms other methods, producing fewer detection errors and more precise edge predictions.

4) Experimental Results on DSIFN-CD Dataset: The DSIFN-CD dataset comprises six different scenes of RS images, including urban and rural areas, which increases the difficulty of CD across various scenarios, making it a highly challenging dataset. The results of various algorithms on the DSIFN-CD dataset are presented in Table III. The table shows that CSTSUNet outperforms all other experimental methods and achieves the highest scores across all five evaluation metrics. Notably, our proposed method obtains scores that are 5.30, 0.60, 3.05, 5.66, and 4.51 points higher than the

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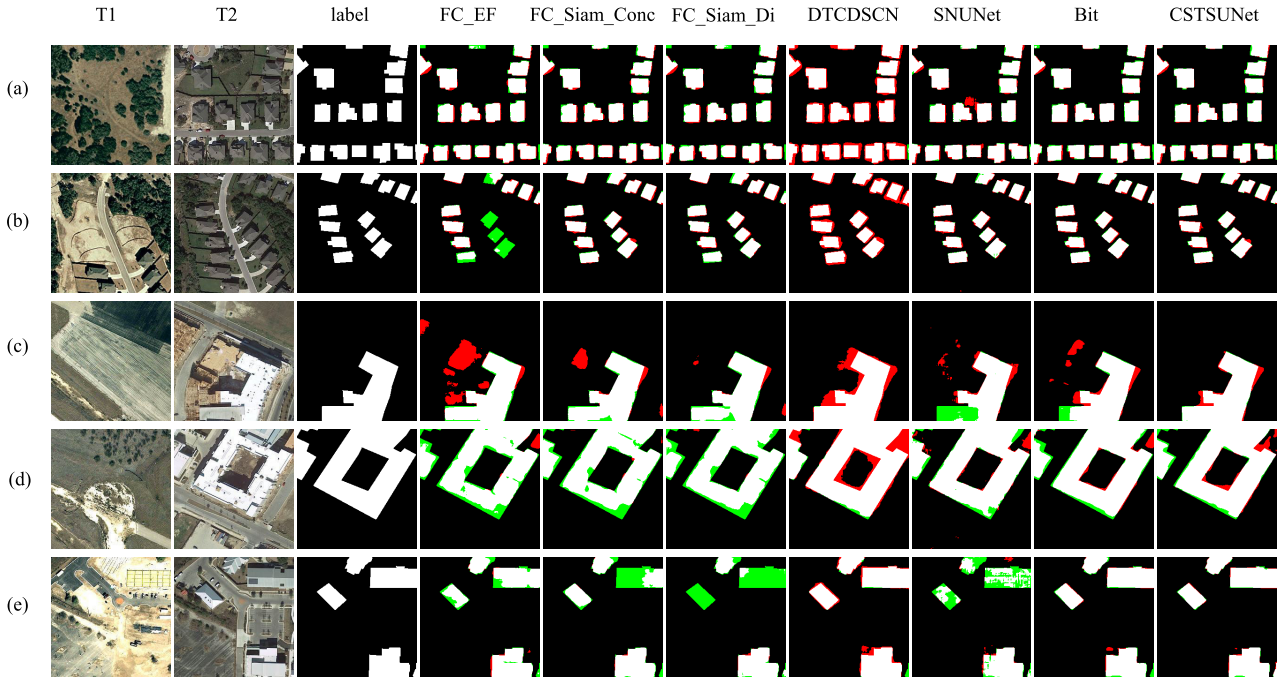


Fig. 4. Visualization results of several CD methods on LEVIR test set. For the convenience of viewing, we use different colors to express different meanings; white means TP, black means TN, red means FP, and green means FN. (a)–(e) Five sets of images randomly selected from the LEVIR test set, respectively.

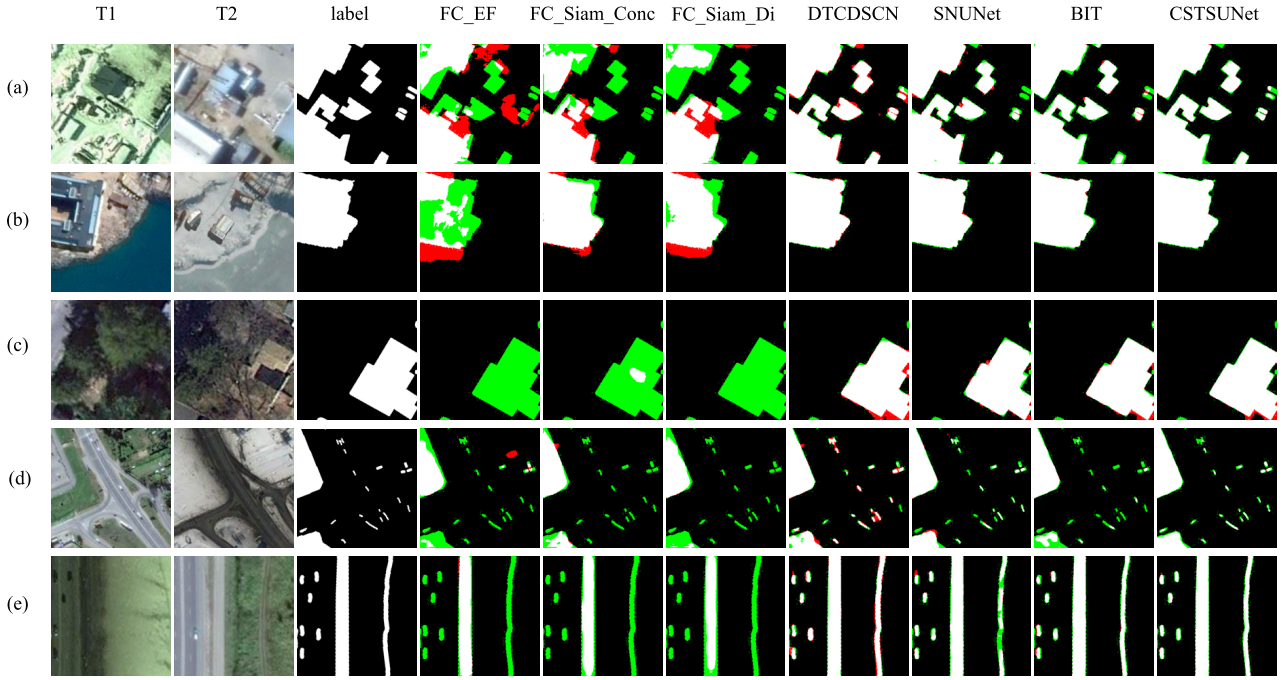


Fig. 5. Visualization results of several CD methods on the CDD test set. The meanings of the different colors are the same as in Fig. 4. (a)–(e) Five sets of images randomly selected from the CDD test set, respectively.

highest scores obtained by other experimental methods for Pre, Rec, $F1$ score, IoU, and OA, respectively. These results demonstrate that, compared to other experimental methods, CSTSUNet exhibits superior performance in CD.

We picked out a few images from the DSIFN-CD test set and applied the same algorithm used in the LEVIR dataset to generate change maps. The performance of these methods on the DSIFN-CD dataset was compared and visualized in Fig. 6. The visualization results provide clear evidence that CSTSUNet outperforms other methods in detecting changes across various scenes.

Fig. 6(a), (b), and (e) depict changes that are caused by the buildings, whereas Fig. 6(c) shows changes arising from the presence of both buildings and roads. The DSIFN-CD dataset is quite challenging due to the complexity of its scenes, and most methods fail to detect changes accurately, with some even producing significant detection errors. Our proposed CSTSUNet method stands out by achieving precise

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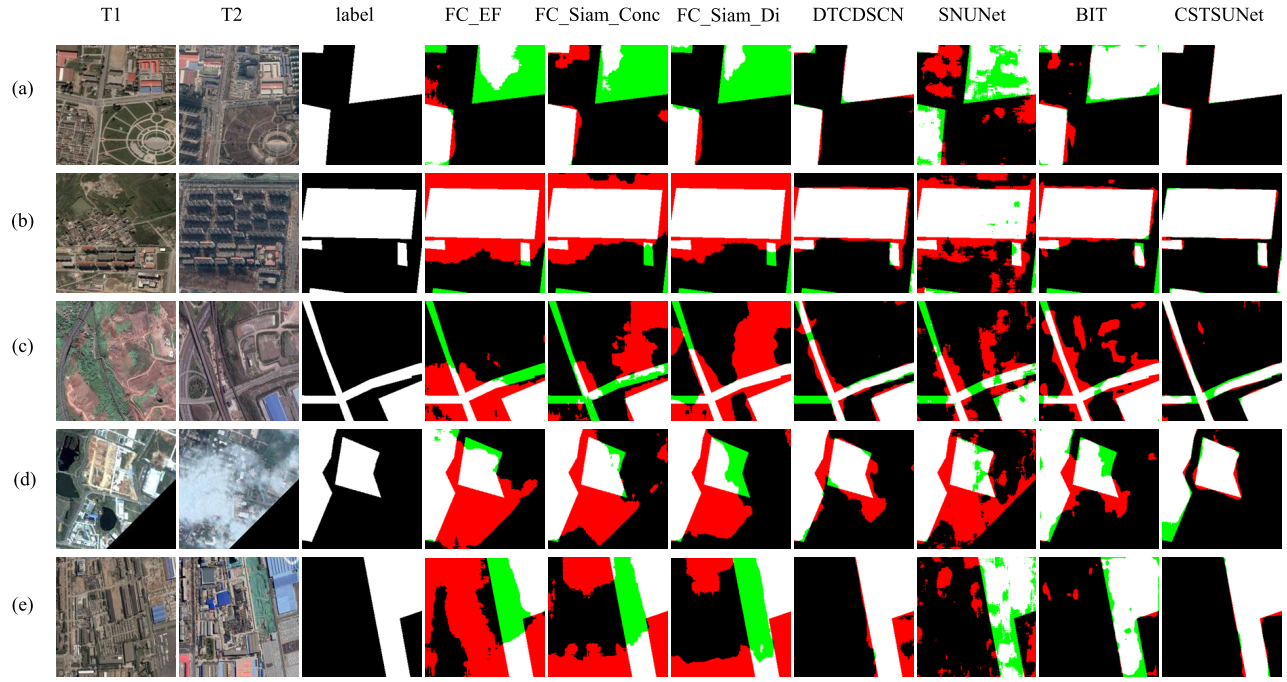


Fig. 6. Visualization results of several CD methods on the DSIFN-CD test set. The meanings of the different colors are the same as in Fig. 4. (a)–(e) Five sets of images randomly selected from the DSIFN-CD test set, respectively.

TABLE III
COMPARISON RESULTS ON THE DSIFN-CD TEST SET. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). COLOR CONVENTION: **BEST**, **SECOND-BEST**

	Pre	Rec	F1	IoU	OA
FC-EF [6]	81.26	77.96	79.58	66.08	85.80
FC-Siam-Conc [6]	81.71	83.37	82.53	70.26	87.48
FC-Siam-Di [6]	84.68	77.04	80.68	67.62	86.91
DTCDSCN [7]	**91.70**	**97.80**	**94.65**	**89.84**	**96.07**
SNUNet [8]	87.35	88.55	87.94	78.48	91.38
BIT [19]	89.21	94.00	91.54	84.40	93.84
CSTSUNet	**97.00**	**98.40**	**97.70**	**95.50**	**98.35**

CD results. In Fig. 6(d), the presence of cloud cover further complicates the CD task, with many existing methods struggling to detect building changes under such conditions. Nevertheless, our CSTSUNet method can accurately detect changes in the presence of cloud occlusion.

E. Model Efficiency and Parameter Analysis

1) Model Efficiency: To ensure a fair comparison of model efficiency, we conducted tests for all methods on a computing server that is equipped with an NVIDIA GeForce RTX 3090 GPU. The results of these tests, including the number of floating-point operations per second (FLOPs), the number of parameters used (Params.), and the Pre/Rec/$F1$/IoU/OA scores for different methods on the LEVIR test sets, are presented in Table IV. In our experiments, we consider the tradeoff between accuracy evaluation and the FLOPs. And in CSTSUnet, the network depth is regarded as an adjustable hyper-parameter to balance accuracy and FLOPs. As we all know, the deeper network requires more calculations, and captures more fine-grained features. The quantitative results listed in Table IV suggest that our

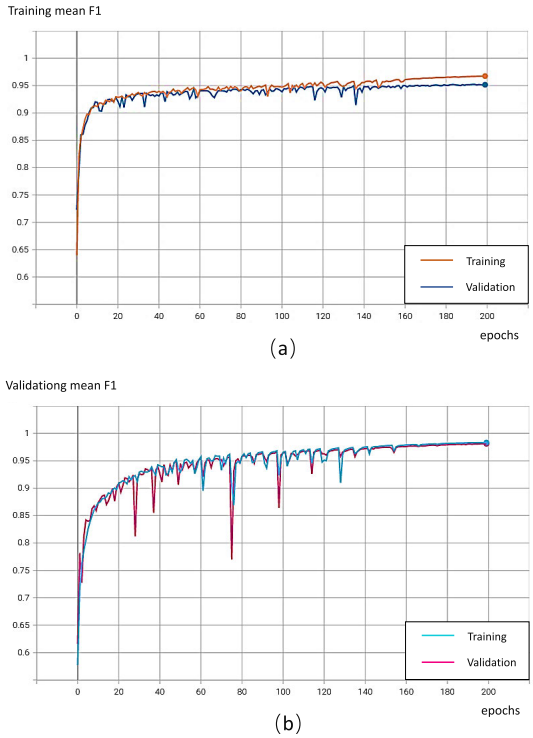


Fig. 7. Accuracy of models for each training epoch. The mean $F1$-score is reported. Training and validation accuracy on (a) LEVIR dataset and (b) CDD dataset.

CSTSUnet can get excellent performance in CD. On LEVIR dataset, the CSTSUnet(ResNet18,stage4) can achieve better performance than other CD methods, and it only has 7.92M parameters and only 12.05G FLOP.

2) Parameter Analysis: Fig. 7 illustrates the mean $F1$-score on the training and validation sets for each training epoch.

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TABLE IV
PERFORMANCE COMPARISON AND ANALYSIS OF MODEL DEPTH ON LEVIR DATASET

	FLOPs(G)	Params(M)	Pre	Rec	F1	IoU	OA
FC-EF [6]	3.58	1.35	88.06	83.70	85.82	75.17	98.59
FC-Siam-Conc [6]	5.33	1.55	91.57	86.04	88.72	79.73	98.89
FC-Siam-Di [6]	4.73	1.35	92.34	83.23	87.55	77.85	98.79
DTCDSCN [7]	13.22	31.26	72.99	96.29	83.03	70.99	98.00
SNUNet [8]	54.83	12.04	89.50	88.87	89.18	80.47	98.90
BIT [19]	10.63	3.50	89.24	89.37	89.31	80.68	98.92
CSTSUNet(Resnet18, stage4)	12.05	7.92	92.57	88.31	90.39	82.47	99.04
CSTSUNet(Resnet18, stage5)	15.43	32.06	92.44	88.98	90.68	82.94	99.07
CSTSUNet(Resnet34, stage4)	16.29	13.31	92.85	88.70	90.73	83.03	99.08
CSTSUNet(Resnet34, stage5)	20.28	42.17	92.51	89.16	90.81	83.16	99.08

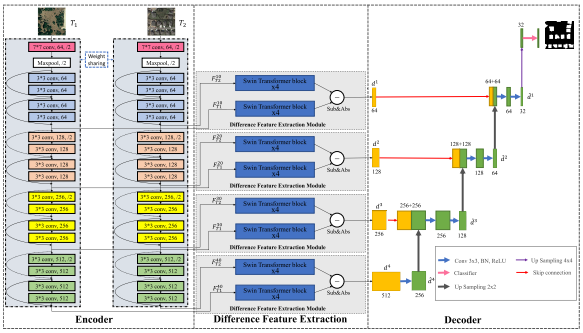


Fig. 8. Overall framework of ablative experiments using Swin transformer method.

Fig. 7(a) shows the training and validation accuracy on the LEVIR dataset, and Fig. 7(b) shows the training and validation accuracy on the CDD dataset. As can be seen from these figures, the model will converge when it reaches a certain epoch. Too much epoch will not significantly increase the CD capability of the model. Therefore, in our experiments, the number of epochs we selected was 200.

F. Ablation Study

It is well known that many factors can affect the model results, such as network structure and parameter initialization method. In this section, we mainly research the influences of Cross Swin transformer, Using Cross Swin transformer at the skip connections of the Unet architecture and model depth on the CSTSUnet model. For these three factors, we conducted ablation experiments on the LEVIR dataset.

1) Impact of Cross Swin Transformer: the Cross Swin transformer is used to perform cross-attention between the two bitemporal images with the same resolution. In experiments, we verify the validity of this block.

The Cross Swin transformer is a modified version of the Swin Transformer block proposed by Shen et al. It replaces the (S)W-MSA module with the (S)W-MCA module, enabling cross-attention between two temporal images instead of self-attention on a single image. To demonstrate its effectiveness, we designed ablative experiments. The overall framework of using the Swin transformer method is shown in Fig. 8. In the difference feature extraction section, we processed the extracted feature maps using the Swin transformer block to obtain enhanced feature representations.

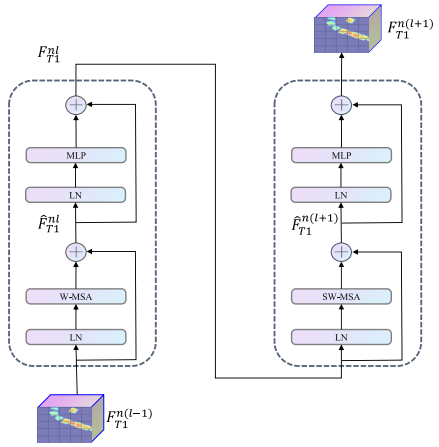


Fig. 9. Two successive Swin transformer blocks, including W-MSA module and SW-MSA module.

TABLE V
COMPARISON RESULTS OF THE CROSS SWIN TRANSFORMER WITH THE METHOD USING THE SWIN TRANSFORMER ON THE LEVIR DATASET

	Pre	Rec	F1	IoU	OA
CSTSUnet(self-attention)	92.96	85.95	89.31	80.69	98.95
CSTSUnet(cross-attention)	92.44	88.98	90.68	82.94	99.07

Two successive Swin transformer blocks, as shown in Fig. 9, including the W-MSA module and SW-MSA module. We then subtracted and took the absolute value of the enhanced feature maps from the two temporal images to obtain the feature changes at the current resolution. Next, we compared the Cross Swin transformer with the method using the Swin transformer on the LEVIR dataset. According to the experimental results presented in Table V, the Cross Swin transformer-based method outperforms the Swin transformer-based method in terms of CD performance.

This demonstrates the effectiveness of our proposed Cross Swin transformer method in handling temporal image data and improving the CD performance compared to the baseline Swin transformer-based approach.

2) Effect of Using Cross Swin Transformer at the Skip Connections of the Unet Architecture: In the past, models using the Unet architecture would take the features extracted from the encoder part of the network and pass them directly to the decoder through skip connections. The decoder would then

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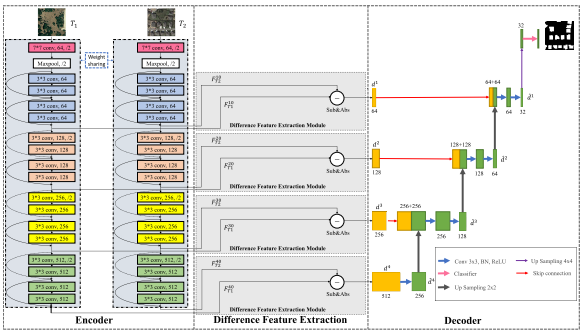


Fig. 10. Overall framework of ablative experiments using the direct connections method.

TABLE VI

COMPARISON RESULTS OF INCORPORATING CROSS SWIN TRANSFORMER AND THE UNET METHOD AT THE SKIP CONNECTIONS ON THE LEVIR DATASET

	Pre	Rec	F1	IoU	OA
the Unet	92.57	86.35	89.35	80.75	98.95
CSTSUnet	92.44	88.98	90.68	82.94	99.07

perform operations such as upsampling, feature fusion, and restoration of the encoder's feature maps to generate an output image with the same resolution as the input image. Fig. 10 illustrates the overall framework of using the direct connections method at the skip connections. However, in CSTSUnet, a Cross Swin transformer is added at the skip connection to process the features extracted from the network encoder before passing them to the decoder. To demonstrate its effectiveness, we compared the integration of Cross Swin transformer at the skip connections with the Unet method on the LEVIR dataset. According to the experimental results presented in Table VI, CSTSUnet outperformed the Unet approach in terms of CD performance.

3) Analysis of Model Depth: Model performance is known to be significantly impacted by its depth. To investigate this relationship, we employed four different depth configurations for our CSTSUnet model: CSTSUnet (ResNet18, stage4), CSTSUnet (ResNet18, stage5), CSTSUnet (ResNet34, stage4), and CSTSUnet (ResNet34, stage5). We used the first four stages and first five stages of the ResNet18 network, as well as the first four stages and first five stages of the ResNet34 network, respectively, as the encoder parts for feature extraction. The performance of these four configurations was evaluated on the LEVIR dataset, and the results are presented in Table IV. As expected, we observed a positive correlation between the depth of the model and its performance. However, as the depth increases, so does the computational cost. Considering both performance and computational efficiency, we selected the CSTSUnet (ResNet18, stage5) configuration as the optimal choice for our experiment.

V. CONCLUSION

In this article, we proposed a new method for CD in RS images using the Cross Swin transformer based Siamese U-shaped network, which named CSTSUNet. The proposed approach effectively tackles the limitations of existing methods that often lead to detection errors due to their high sensitivity to factors such as atmospheric environment, lighting conditions, and phenology.

Our proposed method is firstly based on a hierarchical ResNet with the Siamese U-net structure, which allows parallel processing of bitemporal images and extraction of multiscale features. Then, the difference feature extraction modules of CSTSUNet were mainly composed of the Cross Swin transformer, which allowed interactive communication of features extracted from the bitemporal images at the same spatial resolution. Finally, the decoder generates a reconstructed image that highlights the changes between the two input images, making them more easily interpretable by human observers.

Our experimental results on three different datasets demonstrated that CSTSUNet outperforms other existing methods in terms of accuracy and performance. The proposed method effectively filters out the noise and highlights the changes between the two images, making them more easily interpretable by human observers. The proposed method provides a promising approach for CD in RS images and has the potential to be applied to other fields. In conclusion, the Cross Swin transformer-based Siamese U-shaped network proposed in this article can significantly improve the performance of CD tasks in RS images, and our results provide strong evidence for its effectiveness.

REFERENCES

[1] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[2] P. de Bem, O. de Carvalho Junior, R. F. Guimarães, and R. T. Gomes, "Change detection of deforestation in the Brazilian Amazon using Landsat data and convolutional neural networks," *Remote Sens.*, vol. 12, no. 6, p. 901, Mar. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/6/901

[3] J. Z. Xu, W. Lu, Z. Li, P. Khaitan, and V. Zaytseva, "Building damage detection in satellite imagery using convolutional neural networks," 2019, *arXiv:1910.06444*.

[4] L. I. Kuncheva and W. J. Faithfull, "PCA feature extraction for change detection in multidimensional unlabeled data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 1, pp. 69–80, Jan. 2014.

[5] W. Yao, L. Li, H. Ni, W. Li, and R. Tao, "Hyperspectral anomaly detection based on improved RPCA with non-convex regularization," *Remote Sens.*, vol. 14, no. 6, p. 1343, Mar. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/6/1343

[6] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[7] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[8] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[9] Z. Wang, F. Jiang, T. Liu, F. Xie, and P. Li, "Attention-based spatial and spectral network with PCA-guided self-supervised feature extraction for change detection in hyperspectral images," *Remote Sens.*, vol. 13, no. 23, p. 4927, Dec. 2021. [Online]. Available: https://www.mdpi.com/2072-4292/13/23/4927

[10] M. Gong et al., "A spectral and spatial attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521614.

[11] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel Encoder–Decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.

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[12] D. Meng, F. Gao, J. Dong, Q. Du, and H.-C. Li, "Synthetic aperture radar image change detection via layer attention-based noise-tolerant network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] L. Wang and H. Li, "HMCNet: Hybrid efficient remote sensing images change detection network based on cross-axis attention MLP and CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5236514.

[14] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[15] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[16] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.

[17] F. Song, S. Zhang, T. Lei, Y. Song, and Z. Peng, "MSTDSNet-CD: Multiscale Swin transformer and deeply supervised network for change detection of the fast-growing urban regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[18] J. Ding, X. Li, and L. Zhao, "CDFormer: A hyperspectral image change detection method based on transformer encoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[19] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[20] R. Huang, M. Zhou, Y. Xing, Y. Zou, and W. Fan, "Change detection with various combinations of fluid pyramid integration networks," *Neurocomputing*, vol. 437, pp. 84–94, May 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231221000515

[21] H. R. Kerner, K. L. Wagstaff, B. D. Bue, P. C. Gray, J. F. Bell, and H. Ben Amor, "Toward generalized change detection on planetary surfaces with convolutional autoencoders and transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 3900–3918, Oct. 2019.

[22] W. Zhao, L. Mou, J. Chen, Y. Bo, and W. J. Emery, "Incorporating metric learning and adversarial network for seasonal invariant change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2720–2731, Apr. 2020.

[23] Q. Guo, R. Wang, R. Huang, S. Sun, and Y. Zhang, "IDET: Iterative difference-enhanced transformers for high-quality change detection," 2022, *arXiv:2207.09240*.

[24] X. Song, Z. Hua, and J. Li, "PSTNet: Progressive sampling transformer network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8442–8455, 2022.

[25] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.

[26] D. Mahajan et al., "Exploring the limits of weakly supervised pretraining," 2018, *arXiv:1805.00932*.

[27] H. Touvron, A. Vedaldi, M. Douze, and H. Jégou, "Fixing the train-test resolution discrepancy," 209, *arXiv:1906.06423*.

[28] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[29] W. Liu et al., "SSD: Single shot multibox detector," 2015, *arXiv:1512.02325*.

[30] T. Lin, P. Goyal, R. B. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," 2017, *arXiv:1708.02002*.

[31] L. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," 2018, *arXiv:1802.02611*.

[32] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," 2016, *arXiv:1612.01105*.

[33] A. Kirillov, Y. Wu, K. He, and R. B. Girshick, "PointRend: Image segmentation as rendering," 2019, *arXiv:1912.08193*.

[34] F. Li et al., "Fully convolutional pyramidal networks for semantic segmentation," *IEEE Access*, vol. 8, pp. 229132–229140, 2020.

[35] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," 2015, *arXiv:1505.04597*.

[36] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015, *arXiv:1512.03385*.

[37] T. Chen, Z. Lu, Y. Yang, Y. Zhang, B. Du, and A. Plaza, "A Siamese network based U-Net for change detection in high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2357–2369, 2022.

[38] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016.

[39] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[40] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[41] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[42] H. Chen, C. Wu, B. Du, and L. Zhang, "Deep Siamese multi-scale convolutional network for change detection in multi-temporal VHR images," in *Proc. 10th Int. Workshop Anal. Multitemporal Remote Sens. Images (MultiTemp)*, Aug. 2019, pp. 1–4.

[43] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.

[44] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[45] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," 2021, *arXiv:2103.14030*.

[46] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/10/1662

[47] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. XLII-2, pp. 565–571, May 2018. [Online]. Available: https://isprs-archives.copernicus.org/articles/XLII-2/565/2018/

[48] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[49] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271620301532



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